

MH2100 Calculus III

Midterm Exam 2 (Solutions)

AY 2024/2025, Semester 1

Quantitative Research Society @NTU

October 24, 2024

Overview

- **Coverage:** Lagrange multipliers; extrema on disks; and double integrals.
- **Historical source:** MH2100 Midterm Exam 2, AY2024/2025 Semester 1. Every problem below is transcribed from the official paper. Solutions and marking schemes are unofficial QRS@NTU material.
- **Current administration (AY2026/2027):** Tuesday 6 October 2026, 16:30–18:20 (110 minutes), venue TBA. One double-sided typed A4 sheet is permitted; attachments and sticky notes are not.
- **2026 scope note:** spherical coordinates, general Jacobian transformations, and line integrals are beyond the current midterm scope. They are retained and labelled for historical coverage.

Instructions to candidates (historical, verbatim)

- (1) Do not turn over the exam paper until instructed to do so.
- (2) This midterm exam consists of only three problems, each consisting of two parts.
- (3) Candidates are required to attempt all three problems.
- (4) The point allocation for each problem is shown at the top right-hand corner of its page.
- (5) Candidates may write anywhere on the blank pages reserved for showing work.
- (6) A single double-sided A4-sized help-sheet with only handwritten notes is permitted.

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Question 1

(20 Points)

Problem 1

- (a) Define a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ by

$$f(x, y, z) = y^2 - 10z.$$

Let S denote the sphere with center $(0, 0, 0)$ and radius 6, i.e.,

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 36\}.$$

[10 points] Use the Method of Lagrange Multipliers to find the absolute extrema of $f|_S$.

Remark: $f|_S$ denotes the restriction of f to S .

- (b) Let D denote the closed disk in $\mathbb{R}^2$ with center $(0, 0)$ and radius 1, i.e.,

$$D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}.$$

Define a function $f : D \rightarrow \mathbb{R}$ by

$$f(x, y) = 2x^2 + y^2 - y + 3.$$

- (i) [4 points] Find all the critical points of f , as well as the values of f at these points.

Remark: Any critical point of f must belong to the interior of D .

- (ii) [4 points] Using the standard parametrization $\mathbf{r}$ of the unit circle, given by

$$\forall t \in [0, 2\pi] : \quad \mathbf{r}(t) = (\cos(t), \sin(t)),$$

find the absolute extrema of f on the boundary of D (which is the unit circle).

- (iii) [2 points] Use your work for (b)(i) and (b)(ii) to identify the absolute extrema of f .

Solution

Method 1: Direct calculation

- (a) The Extreme Value Theorem applies because S is compact and f is continuous. With $g = x^2 + y^2 + z^2$, the constraint gradient $\nabla g = 2\langle x, y, z \rangle$ is nonzero on S , and the Lagrange equations are

$$\langle 0, 2y, -10 \rangle = \lambda \langle 2x, 2y, 2z \rangle.$$

The first equation gives either $\lambda = 0$ or $x = 0$; $\lambda = 0$ is impossible because of the third equation. Thus $x = 0$. The second equation gives the branches $y = 0$ or $\lambda = 1$. If $y = 0$, the sphere gives $z = \pm 6$, producing values -60 and 60 . If $\lambda = 1$, the third equation gives $z = -5$, and the sphere gives $y = \pm\sqrt{11}$, producing 61 . Hence

$$\boxed{\max_S f = 61 \text{ at } (0, \sqrt{11}, -5), (0, -\sqrt{11}, -5); \quad \min_S f = -60 \text{ at } (0, 0, 6).}$$

- (b) (i) $\nabla f = \langle 4x, 2y - 1 \rangle$, so the only interior critical point is $(0, 1/2)$, where

$$\boxed{f(0, 1/2) = 11/4.}$$

- (ii) On the boundary, put $x = \cos t$, $y = \sin t$. Then $f = 5 - \sin^2 t - \sin t$. For $s = \sin t \in [-1, 1]$, this is $5 - s^2 - s$, whose maximum is $21/4$ at $s = -1/2$ and minimum is 3 at $s = 1$. Thus

$$\max_{\partial D} f = 21/4 \text{ at } (\sqrt{3}/2, -1/2), (-\sqrt{3}/2, -1/2); \quad \min_{\partial D} f = 3 \text{ at } (0, 1).$$

- (iii) Comparing the interior and boundary values,

$$\min_D f = 11/4 \text{ at } (0, 1/2); \quad \max_D f = 21/4 \text{ at } (\sqrt{3}/2, -1/2), (-\sqrt{3}/2, -1/2).$$

UNOFFICIAL QRS marking scheme (20 points)

- (a) **(10 points)** 2: invokes EVT; 3: Lagrange equations and branches; 3: solves all branches; 2: correct extrema and points.
- (b) **(10 points)**
- (i) **(4 points)** 2: critical equations; 1: point; 1: value.
 - (ii) **(4 points)** 2: boundary parametrisation; 1: extrema; 1: points.
 - (iii) **(2 points)** compares all candidates and states global extrema.

Question 2

(15 Points)

Problem 2

- (a) [5 points] Evaluate

$$\int_0^1 \left[\int_{\sqrt{y}}^1 \frac{1}{\sqrt{x^3+1}} dx \right] dy.$$

Remark: First sketch the region of integration.

- (b) (*beyond 2026 midterm scope*) Let E denote the region in the first octant of $\mathbb{R}^3$ (i.e., $\mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}$) that is
- bounded above by the sphere given by the equation $x^2 + y^2 + z^2 = 1$;
 - bounded below by the closed cone given by the equation $z = \sqrt{\frac{x^2 + y^2}{3}}$.
- (i) [3 points] Describe E in spherical coordinates, using inequalities for ρ , θ , and ϕ .
- (ii) [7 points] Evaluate $\iiint_E 3z \, dV(x, y, z)$ using your work for (b)(i). **Caution:** Work that does not use spherical coordinates will not receive credit.

Solution

Method 1: Direct calculation

- (a) The source region is $0 \leq y \leq 1$ and $\sqrt{y} \leq x \leq 1$, equivalently $0 \leq x \leq 1$ and $0 \leq y \leq x^2$. Hence

$$\int_0^1 \int_0^{x^2} \frac{1}{\sqrt{x^3+1}} dy dx = \int_0^1 \frac{x^2}{\sqrt{x^3+1}} dx.$$

With $w = x^3 + 1$, $dw = 3x^2 dx$, so

$$\boxed{\int_0^1 \frac{x^2}{\sqrt{x^3+1}} dx = \frac{2}{3} (\sqrt{2} - 1)}.$$

- (b) *Beyond 2026 midterm scope.* In the first octant, $0 \leq \theta \leq \pi/2$. The cone has $\tan \phi = \sqrt{3}$, and the region above it has $0 \leq \phi \leq \pi/3$. The sphere gives $0 \leq \rho \leq 1$:

$$\boxed{0 \leq \rho \leq 1, \quad 0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \phi \leq \frac{\pi}{3}}.$$

Therefore

$$\iiint_E 3z \, dV = \int_0^{\pi/2} \int_0^{\pi/3} \int_0^1 3\rho^3 \cos \phi \sin \phi \, d\rho \, d\phi \, d\theta = \boxed{\frac{9\pi}{64}}.$$

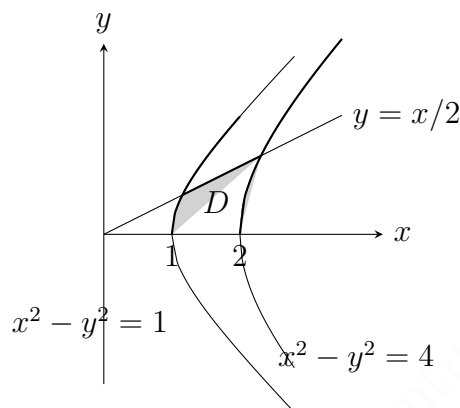
UNOFFICIAL QRS marking scheme (15 points)

- (a) **(5 points)** 2: sketches/reverses the region; 2: integrates in y ; 1: substitutes and obtains $\frac{2}{3}(\sqrt{2} - 1)$.
- (b) **(10 points)**
- (i) **(3 points)** 1: each spherical bound.
- (ii) **(7 points)** 2: correct integrand and Jacobian; 3: integral; 2: value $9\pi/64$.

Question 3*(beyond 2026 midterm scope)***(15 Points)****Problem 3**

(a) Let D denote the region in the first quadrant of $\mathbb{R}^2$ bounded by the following four curves:

- The two hyperbolas given by the equations $x^2 - y^2 = 1$ and $x^2 - y^2 = 4$.
- The two lines given by the equations $y = 0$ and $y = x/2$.



[7 points] Evaluate $\iint_D \frac{y}{x} dA(x, y)$ using a change of variables.

(b) (i) Define a function $f : \mathbb{R}^3 \setminus \{(0, 0, 0)\} \rightarrow \mathbb{R}$ by

$$f(x, y, z) = \frac{1}{x^2 + y^2 + z^2}.$$

Let C denote the curve in $\mathbb{R}^3$ having the parametrization $\mathbf{r} : [0, 2\pi] \rightarrow \mathbb{R}^3$ defined by

$$\mathbf{r}(t) = (\cos t, \sin t, t).$$

[4 points] Evaluate $\int_C f ds$.

(ii) Define a vector field $\mathbf{F}$ on $\mathbb{R}^3$ by

$$\mathbf{F}(x, y, z) = (x, 3xy, -(x + z)).$$

Let C denote the directed line segment going from $(1, 4, 2)$ to $(0, 5, 1)$.

[4 points] Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.

Solution**Method 1: Direct calculation**

(a) *Beyond 2026 midterm scope.* Use $u = x^2 - y^2$ and $v = y/x$. The boundaries become $1 \leq u \leq 4$ and $0 \leq v \leq 1/2$, while

$$\frac{\partial(u, v)}{\partial(x, y)} = 2(1 - v^2), \quad dA = \frac{du dv}{2(1 - v^2)}.$$

On the positive branch in the first quadrant, the inverse is $x = \sqrt{u/(1 - v^2)}$, $y = vx$, so the transformation is one-to-one on D . Hence

$$\iint_D \frac{y}{x} dA = \int_1^4 \int_0^{1/2} \frac{v}{2(1 - v^2)} dv du = \boxed{\frac{3}{4} \ln \left(\frac{4}{3} \right)}.$$

(b) *Beyond 2026 midterm scope.*

(i) $\|\mathbf{r}'(t)\| = \sqrt{2}$ and $f(\mathbf{r}(t)) = 1/(1+t^2)$. Thus

$$\int_C f \, ds = \sqrt{2} \int_0^{2\pi} \frac{dt}{1+t^2} = \sqrt{2} \arctan(2\pi).$$

(ii) Put $\mathbf{r}(t) = \langle 1-t, 4+t, 2-t \rangle$, $0 \leq t \leq 1$. Then

$$\mathbf{F}(\mathbf{r}(t)) \cdot \langle -1, 1, -1 \rangle = 14 - 10t - 3t^2.$$

Therefore

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 (14 - 10t - 3t^2) \, dt = \boxed{8}.$$

UNOFFICIAL QRS marking scheme (15 points)

(a) **(7 points)** 2: valid transformation; 2: maps region; 2: Jacobian; 1: value.

(b) **(8 points)**

(i) **(4 points)** 1: parametrisation; 1: arc length; 1: integral; 1: value.

(ii) **(4 points)** 1: segment parametrisation; 2: dot product; 1: value 8.